|| Shri Hari ||

# AN OVERVIEW OF DATABASE MANAGEMENT SYSTEM (DBMS)

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## Introduction

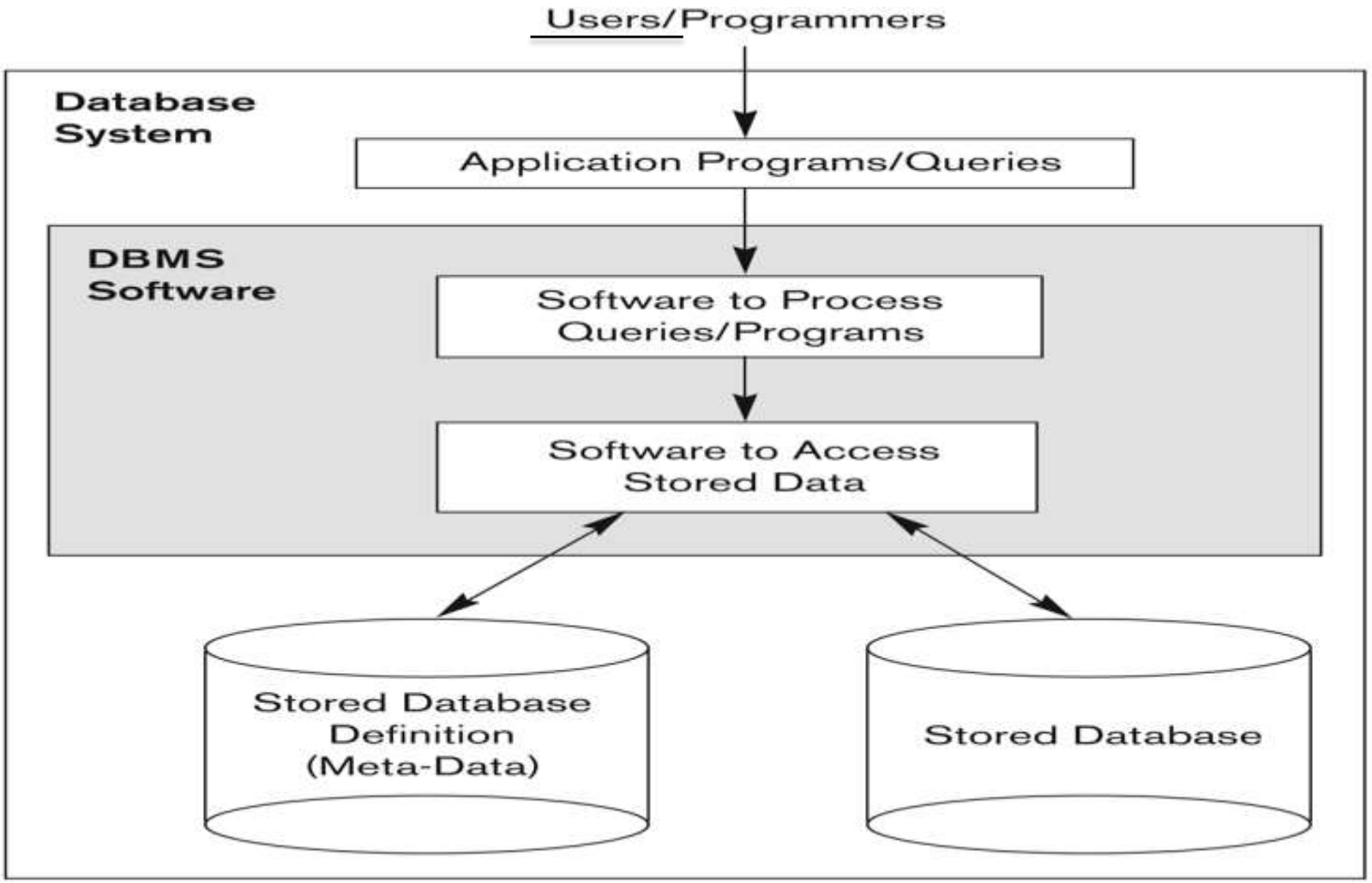
# What is a DBMS?



- A large, integrated collection of data
- Models a real-world *enterprise*
  - *Entities* (e.g., Students, Courses)
  - *Relationships* (e.g., Alice is enrolled in 145)

A **Database Management System (DBMS)** is a piece of software designed to store and manage databases

# Simplified database system environment



# What you will learn in this Lecture

- Overview of DBMS
- Need for DBMS
- Types of Databases
- Components of DBMS
- Application of DBMS
- Advantages of DBMS
- Disadvantages of DBMS

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# Basic Definitions

- "***Data***" – are raw facts

-It must be formatted for storage, processing, and presentation.

- "***Database***" –collection of logically interrelated data
- "***Database Management***"-the creation & maintenance of a collection of organized data.
- "***Database Management System (DBMS)*** " - It is a software package designed to *define*, manipulate, retrieve and manage data in a database.

# Example of a Database

- **Mini-world for the example:**
  - Part of a UNIVERSITY environment.
- **Some mini-world *entities*:**
  - STUDENTs
  - COURSEs
  - SECTIONs (of COURSEs)
  - (academic) DEPARTMENTs
  - INSTRUCTORs

# Example of a Database (cont'd.)

- **Some mini-world *relationships*:**
  - SECTIONs *are of specific* COURSEs
  - STUDENTs *take* SECTIONs
  - COURSEs *have prerequisite* COURSEs
  - INSTRUCTORs *teach* SECTIONs
  - COURSEs *are offered by* DEPARTMENTs
  - STUDENTs *major in* DEPARTMENTs

- Note: The above entities and relationships are typically expressed in a conceptual data model, such as the ENTITY-RELATIONSHIP data model

# Example of a simple database

**COURSE**

| Course_name | Course_number | Credit_hours | Department |
|---|---|---|---|
| Intro to Computer Science | CS1310 | 4 | CS |
| Data Structures | CS3320 | 4 | CS |
| Discrete Mathematics | MATH2410 | 3 | MATH |
| Database | CS3380 | 3 | CS |

**SECTION**

| Section_identifier | Course_number | Semester | Year | Instructor |
|---|---|---|---|---|
| 85 | MATH2410 | Fall | 04 | King |
| 92 | CS1310 | Fall | 04 | Anderson |
| 102 | CS3320 | Spring | 05 | Knuth |
| 112 | MATH2410 | Fall | 05 | Chang |
| 119 | CS1310 | Fall | 05 | Anderson |
| 135 | CS3380 | Fall | 05 | Stone |

**GRADE_REPORT**

| Student_number | Section_identifier | Grade |
|---|---|---|
| 17 | 112 | B |
| 17 | 119 | C |
| 8 | 85 | A |
| 8 | 92 | A |
| 8 | 102 | B |
| 8 | 135 | A |

**PREREQUISITE**

| Course_number | Prerequisite_number |
|---|---|
| CS3380 | CS3320 |
| CS3380 | MATH2410 |
| CS3320 | CS1310 |

# Example of a simplified database catalog

**RELATIONS**

| Relation_name | No_of_columns |
|---|---|
| STUDENT | 4 |
| COURSE | 4 |
| SECTION | 5 |
| GRADE_REPORT | 3 |
| PREREQUISITE | 2 |

**COLUMNS**

| Column_name | Data_type | Belongs_to_relation |
|---|---|---|
| Name | Character (30) | STUDENT |
| Student_number | Character (4) | STUDENT |
| Class | Integer (1) | STUDENT |
| Major | Major_type | STUDENT |
| Course_name | Character (10) | COURSE |
| Course_number | XXXXNNNN | COURSE |
| .... | .... | ..... |
| .... | .... | ..... |
| .... | .... | ..... |
| Prerequisite_number | XXXXNNNN | PREREQUISITE |

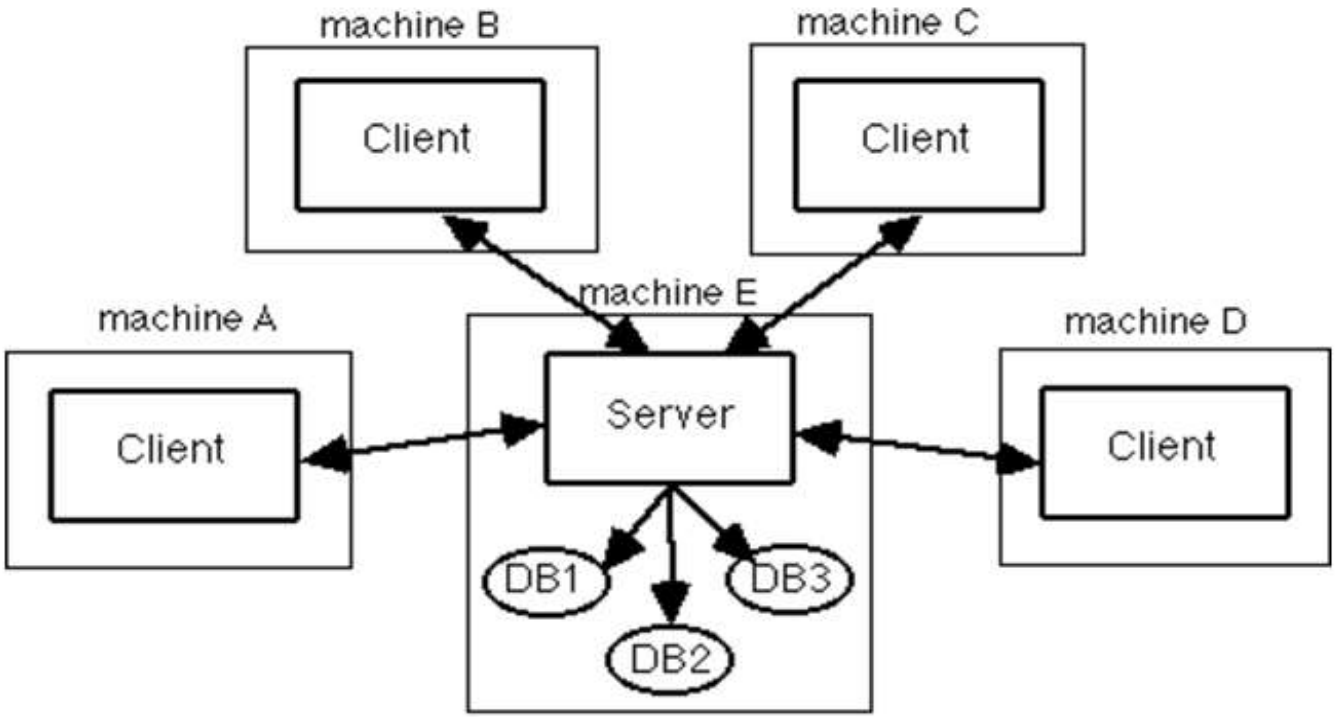
*Note*: Major_type is defined as an enumerared type with all known majors. XXXXNNNN is used to define a type with four alpha characters followed by four digits

# Need for DBMS

- Provide a highly efficient method for handling large amount of different types of data with ease.
- Database allows data to be stored systematically.
- Data can be easily retrieved, filtered, sorted and updated efficiently and accurately.

# Types of Databases

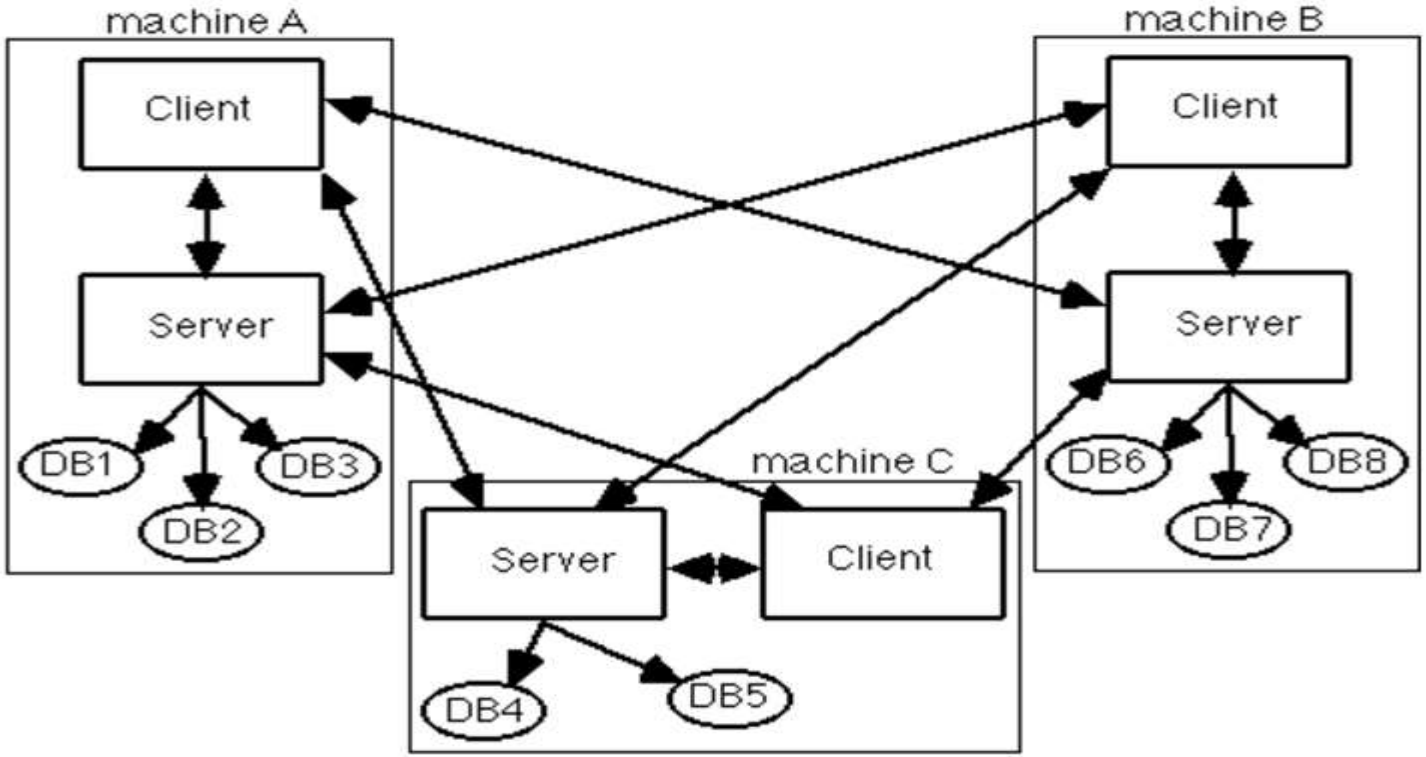
1. Centralized Database:




**Centralized Databases**

# Based on Database location(s)

2. Distributed Database:

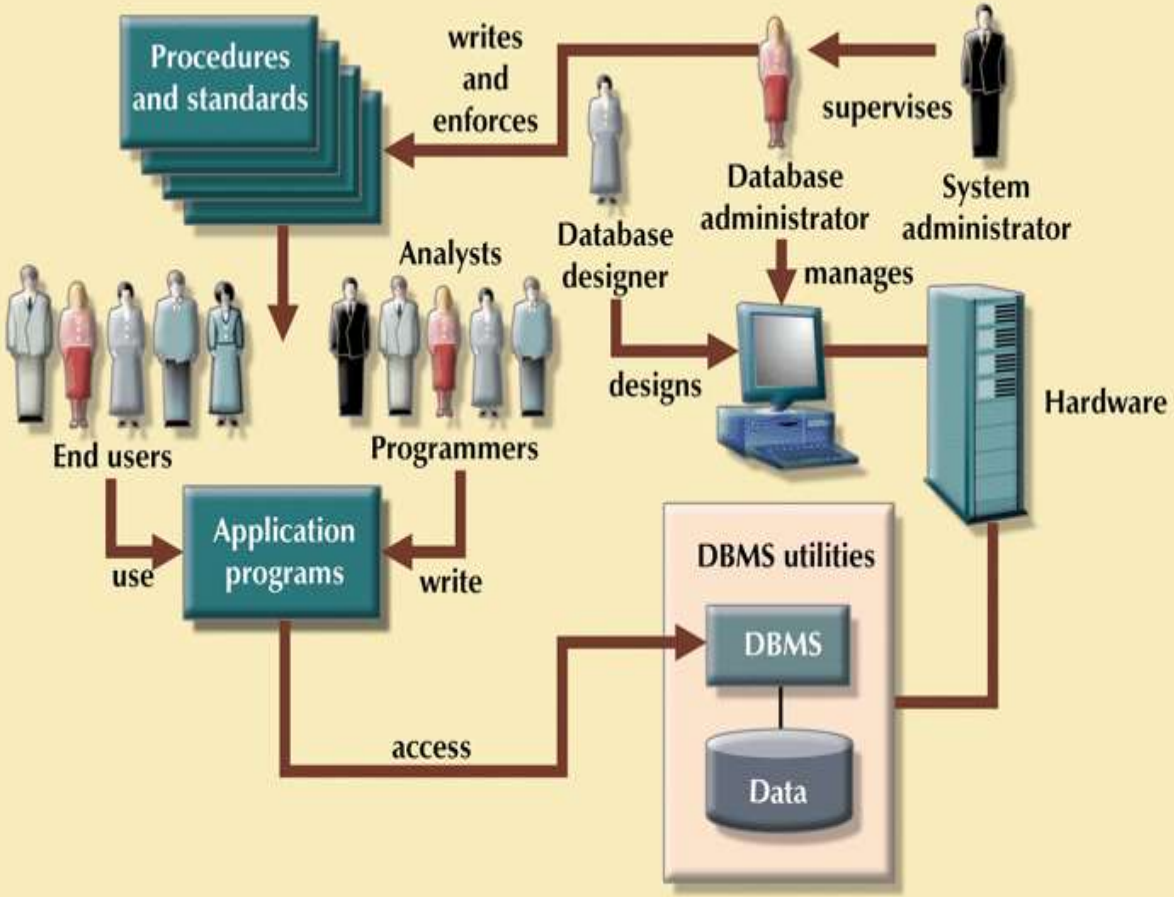



**Distributed Databases**

# Components of DBMS

1. Database Users
2. Data
3. Software and Procedures
4. Hardware
5. Database Access Language

# The database system environment



# 1. Database Users

- **End-users:** They use the data for queries, reports and some of them update the database content. End-users can be categorized into:
  - **Sophisticated:** These include business analysts, scientists, engineers, others thoroughly familiar with the system capabilities.
  - **Naive** or Parametric: They use previously well-defined functions against the database.

    Examples are bank-tellers or reservation clerks who do this activity for an entire shift of operations using standard types of queries.

# Database Users (cont'd.)

- **Database administrators:**

Responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources, controlling its use and monitoring efficiency of operations.

- **Application programmers(Software Engineers):**

Responsible for writing database application programs in some programming language.

# 2. Data

- The collection of facts stored in the database
- data is stored, updated and retrieved to from a database.
- Database contains both operational data and metadata.

# 3. Software

- Operating System software
- DBMS software
- Network Software
- Application Programs and utility software
- Procedures: instructions and rules that govern the design and use of the database system.

# 4. Hardware

- PC
- Network of computers
- Various Storage devices
- Input devices
- Output Devices

# 5. Database Languages

- Data Definition Language (DDL)

  Create, Alter, Drop

- Data Manipulation Language (DML)

  Insert, Select, Update, Delete

- Data Control Language (DCL)

  Grant, Revoke

- Transaction Control Language (TCL)

  Commit, Rollback.

# Application of DBMS

- Banking

- Airline

- Universities

- Credit and Transactions

- Sales

- Human resource

# Application of DBMS (cont'd.)

- **Traditional Applications:**
  - Numeric and Textual Databases

- **More Recent Applications:**
  - Multimedia Databases
  - Geographic Information Systems (GIS)
  - Data Warehouses
  - Real-time and Active Databases
  - Many other applications

# Advantages of Using the Database Approach

- Controlling redundancy in data storage and in development and maintenance efforts.
  - Sharing of data among multiple users.
- Restricting unauthorized access to data.
- Providing Storage Structures (e.g. indexes) for efficient Query Processing.
- Providing backup and recovery services.
- Providing multiple interfaces to different classes of users.
- Representing complex relationships among data.
- Enforcing integrity constraints on the database.

# When not to use a DBMS

- If the database and applications are simple, well defined, and not expected to change.
- High initial investment and possible need for additional hardware.
- Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- If access to data by multiple users is not required.

# What are the current trends ??

- Multimedia Databases
- Distributed Database
- Document-oriented Databases
- Mobile & embedded Databases

# Thank You